Venkata Amith Palacherla

Lead Perception Engineer | Robot Perception | Computer Vision | SLAM | Autonomous Systems

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Professional Summary

Lead Perception Engineer specializing in robotic perception, 3D computer vision, SLAM, localization, and multi-sensor systems for autonomous robots. Experience leading the development and deployment of real-time perception software for **industrial robotic manipulators and autonomous mobile robots** in commercial production environments. Research background in semantic SLAM, active localization, autonomous navigation, state estimation, and learning-based perception, with peer-reviewed publications in **IEEE Robotics and Automation Letters (RA-L)** and ICARA and a granted robotics patent.

Research & Review Expertise

Robot Perception • Computer Vision • 3D Perception • Semantic SLAM • Localization & Mapping • Multi-Sensor Fusion • State Estimation • Autonomous Navigation • Object Tracking • Learning-Based Perception • Industrial Robotics

Selected Publications

- **Palacherla, V.A.** et al., "Floor Plan Based Active Global Localization and Navigation Aid for Persons with Blindness and Low Vision," **IEEE Robotics and Automation Letters (RA-L)**, 2024; presented at **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)**.
- **Palacherla, V.A.** et al., "Efficient Real-Time Localization in Prior Indoor Maps Using Semantic SLAM," **9th International Conference on Automation, Robotics and Applications (ICARA)**, 2023.

Professional Experience

Cohesive Robotics **Jun 2024 – Present; Jun 2023 – Aug 2023**
Lead Perception Engineer *New York, NY*

- **Lead the perception and computer vision technical effort** for industrial robotic systems, architecting real-time **3D perception and spatial reasoning** capabilities in C++ and ROS 2 for autonomous manufacturing processes.
- Develop perception algorithms that transform RGB-D, stereo, laser-profile, and point-cloud sensor data into reliable geometric representations used by downstream robot planning, control, and execution systems.
- Developed automated **multi-sensor spatial and hand-eye calibration** capabilities for robotic manipulation systems, improving geometric consistency and repeatability during production operation.
- Implemented real-time **3D point cloud filtering, surface segmentation, geometric feature extraction, and spatial reasoning** using PCL and Open3D for autonomous manipulation of complex manufactured parts.
- Designed **modular perception APIs and software interfaces** integrating computer vision capabilities with robotic planning, control, and process-execution systems.
- Built **perception telemetry, validation, and regression-testing infrastructure** to reproduce field failures, benchmark algorithm performance, and improve reliability of deployed robotic systems.
- **Led deployment and commissioning** of perception software across **2 commercial robotic workcells** operating in daily manufacturing production.

Skylla Technologies / Movensys Corp. **Sep 2023 – May 2025**
Robotics Software Engineer – Perception & Navigation *Boston, MA*

- Developed multi-sensor perception and state-estimation modules for autonomous mobile robots (AMRs), fusing **2D/3D LiDAR, wheel odometry, IMU, and vehicle-state measurements** using Extended Kalman Filtering (EKF).
- Developed **object tracking and persistence** capabilities including data association, track lifecycle management, and occlusion handling for dynamic environments shared by multiple autonomous robots.
- Engineered **camera-based map validation and fiducial verification tools** to identify survey and localization-reference errors, achieving sub-centimeter 2D map alignment.
- Designed a **wheel-drift and slip detection system** using motor feedback and vehicle kinematics to identify abnormal motion and mitigate localization divergence under heavy-load operating conditions.
- Led on-site sensor bring-up, cross-calibration, perception validation, and localization tuning for an **8-AMR autonomous fleet in Japan**, resolving spatial drift and supporting **$4M client acceptance**.

Control/Robotics Research Laboratory, New York University **Jan 2022 – May 2023**
Graduate Research Assistant – Computer Vision & SLAM *New York, NY*

- Researched and developed a real-time **graph-based Semantic SLAM and active localization system in C++**, combining visual-inertial odometry, semantic observations, and structural prior floor plans for localization in GPS-denied indoor environments.
- Developed semantic localization and navigation methods involving hypothesis generation, particle clustering and scoring, localization confirmation, and autonomous navigation toward informative observations.
- Integrated deep-learning semantic segmentation with **3D geometric and voxel-based filtering** on embedded NVIDIA Jetson platforms.
- Optimized perception-model inference using **TensorRT FP16 quantization**, increasing real-time vision update rates by 10 Hz and improving 3D spatial fidelity by 50%.
- Achieved **90% localization recovery** and **0.1 m tracking accuracy** in real-world benchmark evaluations by combining active visual localization with structural floor-plan constraints.

Shiv Nadar University **Aug 2020 – Jul 2021**
Robotics Systems Research Assistant *Delhi, India*

- Developed perception, tracking, and obstacle-classification software for **autonomous agricultural machinery**, integrating LiDAR, stereo vision, GNSS, and vehicle-state information under dust-degraded outdoor conditions.
- Co-invented an autonomous machine perception, state-estimation, and actuation architecture resulting in **Indian Patent No. 475111**, subsequently licensed to FarmON Agristack.
- Directed a **10-member multidisciplinary engineering team** developing autonomous agricultural robotics technologies and secured **$15,000 in competitive national technical grants**.

Patents

- **Palacherla, V.A.** et al., "Method for Making Bale Using an Autonomous Baler," **Indian Patent No. 475111**, granted 2023.

Technical Expertise

3D Computer Vision & Perception: 3D Point Cloud Processing (PCL, Open3D), 3D Laser Profilers, RGB-D & Stereo Vision, Semantic Segmentation, Surface Reconstruction, Geometric Feature Extraction, Object Tracking, OpenCV, PyTorch

Localization, Navigation & State Estimation: Multi-Sensor Fusion, Extended Kalman Filtering (EKF), Graph-Based SLAM, Semantic SLAM, 2D/3D Occupancy Grids, Dynamic Obstacle Tracking, Autonomous Navigation, AMRs

Robot Calibration & Manipulation: Multi-Sensor Spatial Calibration, Hand-Eye Calibration, Industrial Robotic Manipulators, Sensor-to-Robot Registration, Robot Workcell Integration

Software & Deployment: Modern C++ (C++14/17/20), Python, ROS / ROS 2, Eigen, TensorRT, CUDA, Linux, CMake, Docker, Git, Hardware-in-the-Loop Testing, Simulation, Telemetry, Regression Testing

Education

New York University **May 2023**
Master of Science in Mechatronics and Robotics *New York, NY*

Shiv Nadar University **May 2020**
Bachelor of Technology in Mechanical Engineering *Delhi, India*